

High-pressure processing of foods: a narrative review of lethality, injury, and quality when heat is not the kill step

Abstract

Heat remains the default way to make food microbiologically safe because it can deliver large, predictable reductions of vegetative pathogens. The same heat also drives flavour loss, enzyme damage, and protein denaturation that consumers and some product classes cannot tolerate. High-pressure processing (HPP) applies hundreds of megapascals, usually at chilled or ambient temperature, through a packed food so that pressure is uniform regardless of piece size. Vegetative cells often die; bacterial spores and several enzymes often do not. This narrative review follows those tensions as one argument rather than as a single log-reduction number. Matrix, solute, recovery medium, and injury change the kill that a given pressure appears to deliver; the same cycle may still fail legal or 5-log pasteurisation criteria in milk and juices; and quality (texture, polyphenol oxidase, bioactive proteins, allergenicity) often moves against plate counts rather than with them. Primary evidence is laboratory challenges, storage trials, one scientific opinion, and systematic or scoping reviews; almost none of it can be pooled. Those patterns do not imply that HPP is a sterilant, that it is equivalent to high-temperature short-time milk pasteurisation, or that every nonthermal field (ultrasound, plasma, irradiation) can be substituted for pressure. The next measurements that would change practice are paired pathogen and protein assays in milk, juice validations that name *Escherichia coli* O157:H7 or *Salmonella* rather than aerobic plate count, and spore kinetics recorded with in-situ temperature under pressure.

Keywords

high-pressure processing; high-pressure thermal processing; microbial inactivation; sublethal injury; milk pasteurisation; fruit juice; ready-to-eat meat

Key Summary Points

- Vegetative kill under high pressure is real, but the log reduction is not a constant read from the set-point; matrix, solute, recovery medium, and injury move it (laboratory challenges and one scoping review of ready-to-eat meats).
- Legal milk pasteurisation is not interchangeable with 400–600 MPa on the included scientific opinion.

- A 5-log drop in aerobic plate count is not a named-pathogen juice validation.
- Enzymes, bioactive proteins, and texture often move against plate counts on the same cycle, so they have to be named as endpoints.
- Spore tails remain at commercial pressures; either heat under pressure or a planned cold chain still has to carry that control.

1. Introduction

Foodborne illness remains a large-scale problem. Fruit-and-vegetable HPP reviews reciting World Health Organization figures attribute hundreds of millions of illnesses and hundreds of thousands of deaths each year to contaminated food, with vegetative bacteria, spores, parasites, and viruses as the usual agents [1]. Thermal pasteurisation and sterilisation remain the default controls because time–temperature combinations can be written into law. In the European Union, milk pasteurisation is at least 72 °C for 15 s or 63 °C for 30 min, with a 5 log₁₀ performance criterion as the international reference for several milk pathogens [2]. Juice processors in the United States are told to achieve a 5-log reduction of the pertinent pathogen [3]. Heat meets those specifications in most high-water-activity foods. It also cooks colour, volatiles, and whey proteins that some products are sold for.

HPP, also called high hydrostatic pressure, is the industrial attempt to keep that specification without the cook. The packed food is loaded into a water-filled vessel. More water is pumped in until the pack sits under isostatic pressure, typically 300–600 MPa for seconds to minutes. After the hold, the extra water is released and the baskets are unloaded [4]. Because pressure is transmitted rapidly and uniformly, every point in a juice pouch or a ham slice sees the same set-point at the same time. Piece size therefore does not create a cold centre the way conduction heating does [2]. Milk had already been pressurised in 1899, yet commercial lines only appeared in the 1990s, once vessels and cycle times (about six minutes for a competitive 600 MPa hold by 2019) made a refrigerated cold-pasteurisation product economically plausible [4]. Processors use the method where a second heat cook would destroy the product they are selling. They do not use it as a drop-in sterilant. HPP alone does not inactivate most microbial spores or many enzymes [4].

The measured kill is not a constant read from the gauge. Included reviews compile membrane leakage, lipid-bilayer phase change, ribosomal subunit dissociation, and aggregation of higher-order protein structure without cleavage of the peptide chain as contributing routes under pressure [5,6]. Those routes are consistent with damage to non-covalent assemblies rather than to primary sequence. They are not a single primary experiment that proves one molecular cause for every log reduction.

Vegetative inactivation still depends on intrinsic food properties (water activity and pH), extrinsic settings (pressure and holding time, with a threshold below which little happens), and the organism (taxon, strain, and physiological state, so that stationary-phase cells are harder than exponential-phase cells) [1,2]. Gram-negative rods are often more sensitive than Gram-positive cocci. Yeasts and moulds are often easier than bacteria. Bacterial spores are harder than either. A given matrix can reorder that list [1,2,7]. Some *Bacillus subtilis* spores have been reported to survive 1200 MPa at ambient temperature [1]. Commercial vessels rarely exceed about 600 MPa without an added heat supply. Surviving spores after a cold cycle are therefore managed by refrigeration below 7 °C rather than by a retort-like sterility claim [1,4].

The quality claim follows from the same physics. Because the bulk temperature stays low, Maillard browning and caramelisation are largely avoided, so vitamins, colour, and “fresh-like” flavour can survive better than after a heat exchanger [5]. That claim is still product-specific. Enzymes and gels are different molecular targets from vegetative membranes. Polyphenol oxidase (PPO) in grape juice can remain largely active after a pressure cycle that already cuts aerobic counts [8]. Donkey-milk lysozyme and lactoferrin lose less activity at 400 MPa than at 75 °C, but 500 MPa starts to spend that advantage [9]. Brown kelp hardness falls by tens of percent as pressure rises from 200 to 600 MPa, after the same industrial window that is sold as gentle for juices [10].

Ready-to-eat (RTE) cooked meats are a commercial HPP success for *Listeria monocytogenes* because a post-lethality pressure cycle can replace a second cook. Reported log reductions at the same 500–600 MPa window still span 1–7 log colony-forming units per gram (CFU/g), and many trials used a single strain [7]. Commercial use therefore does not make the log a constant.

Two regulatory questions sit under that commercial success, and both exist because a vegetative log is not the legal object. One is whether HPP milk is equivalent to legal thermal pasteurisation. The European Food Safety Authority (EFSA) biological hazards panel modelled industrial 600 MPa for 6 min and concluded that, for the milk pathogens that could be compared, HPP delivered fewer log₁₀ reductions than the legal heat treatments [2]. The industrial cycle is not a substitute on that metric. Alkaline phosphatase, the heat-pasteurisation indicator, is relatively pressure-resistant and would at best flag overprocessing [2]. A passing alkaline phosphatase test cannot verify an HPP milk line the way it verifies a heat exchanger.

The other question is whether juice HPP cycles that report a 5-log drop in aerobic plate count (APC) satisfy a named-pathogen HACCP plan. Several included juice papers never inoculated *E. coli* O157:H7 or *Salmonella*, although the US rule is written for the pertinent pathogen rather than for background aerobes [3,11].

A third problem is injury. A blank plate the afternoon of treatment is not the same as death. After HPP, membrane pores can later shrink [12], survivors can hide on selective agar [13], and injured cells can grow back in storage [14,15]. Culture methods can therefore overestimate kill when viable but nonculturable cells remain. Cited HPP work has shown an 8-log drop in culturable *L. monocytogenes* and *Salmonella* Typhimurium while about 4 log still stained as viable, and a >7-log culture reduction of *S. Typhimurium* against a 0.72-log drop by propidium-iodide-excluding flow cytometry [5].

High-pressure thermal processing (HPTP) adds heat under pressure to finish spores. That addition costs the “nonthermal” label. Adiabatic compression heating (about 3 °C per 100 MPa in water-rich foods, more in fats) plus decompression cooling can put the product above 100 °C for far less time than a retort [4,16]. Dedicated industrial HPTP lines were still not a retail reality in the included engineering review [16]. Most commercial 600 MPa plants therefore still sell a refrigerated vegetative hurdle rather than a shelf-stable spore cook. Hurdles (freezing before pressure, ultrasound, plasma-activated water, plant extracts) try to buy extra vegetative lethality without that full cook [14,17,18,19,20,21].

The aim of this review is to compare included studies on HPP and related nonthermal processes as microbial and quality technologies. The central argument of this review is that pressure is a reliable vegetative hurdle whose log reduction cannot be read from the set-point alone, while quality preservation is an endpoint that must be named (enzyme, protein, texture, allergen) rather than assumed.

2. Methods

2.1 Search and sources

Records came from a Scopus BibTeX export dated 20 September 2026 (359 bibliographic records, 2021–2026). Public full texts were taken only when a PDF body or Europe PMC JATS XML could be opened without a subscription. One additional plant-food HPP review was retrieved after an OpenAlex search aimed at spore kinetics [4].

2.2 Eligibility

Peer-reviewed journal articles and reviews whose titles named HPP/HHP/HPH, ultrasound, pulsed electric fields, plasma, irradiation, or nonthermal processing in a human-food, juice, meat, dairy, produce, or inactivation context. Books, chapters, conference papers, retracted items, and pet-food titles were excluded. Full-text inclusion required a readable public file of the named paper.

2.3 Study selection

After title screening, 182 records were sought. Forty-nine public files were obtained; six were excluded (wrong document, unreadable stub, bibliometric-only, off-topic production or process-control reviews). Forty-three export full texts plus one gap-fill review were included ($n = 44$). Designs range from laboratory inoculations to an EFSA opinion; they are not a single evidence grade. Screening counts live beside the manuscript, not in the Discussion. Extraction used claim-ready facts from each full text (pressure, time, organism, matrix, endpoint, and what the design cannot show). Incommensurable rows were kept in one literature table and then curated into Tables 1–3 of this article rather than averaged. Primary endpoints in those tables are named explicitly in the thematic sections that follow.

3. Pressure, membranes, and why the same megapascals do not give the same kill

Vegetative lethality under HPP is often large, but the log reduction is a family of outcomes rather than one number on a gauge. Ice, solute, strain, and recovery medium change what a given megapascal appears to do. Ice is treated first because it changes the physics of the cell wall rather than adding a second unit operation in name only.

3.1 Freezing before pressure

Freezing before pressure can convert a mild cycle into a large reduction for a Gram-negative surrogate, but the extra kill is mechanical rather than a new legal pasteurisation process. In phosphate buffer, *E. coli* K12 lost 1.83 log CFU·ml⁻¹ after 400 MPa for 9 min at 4 °C when the sample had been held at 4 °C [18]. The same pressure–time after freezing at –24 °C or –80 °C produced 5.63 and 6.83 log reductions [18]. Scanning electron micrographs showed wall rupture only in the frozen, pressurised cells [18]. Orange juice at 250 MPa for 15 min stayed at or below 0.42 log kill at pH 4.5 and 5.8 unless it was frozen first, when reductions rose to about 4 log [18]. After freezing, pH mattered less than it had in the unfrozen juice. Freeze–thaw without pressure already leaked nucleic acids and killed about 1.5–1.7 log in PBS [18]. Pressurising the frozen samples drove supernatant DNA/RNA much higher [18]. A 1 s hold at 300 MPa killed 2.6–3.5 log in frozen PBS versus 0.01 log when unfrozen [18]. The authors attribute the extra logs to ice I being driven toward denser polymorphs (a volume drop near 315 MPa) while the wall is rigid [18]. That explanation is consistent with the SEM and leakage data. It was not mapped as a complete ice-phase diagram in the same paper. The process that delivered those extra logs is therefore freeze-then-pressure, not HPP alone.

3.2 Strain and solute in ready-to-eat meat

An ingredient already in a cooked-ham recipe can change HPP kill without changing the set-point. Processors add lactate as a flavour salt and as a *Listeria* growth control. After 400 MPa, that same solute reduced inactivation in a dose-dependent way. At 10 min, 1.4% lactate cut kill by 0.5–1.46 log versus the control depending on strain, and 2.8% lactate widened that gap [22]. Product pH and water activity did not change, so the protection is not explained by a trivial aw drop in this matrix [22]. Curve shape stayed strain-specific (shoulder, near-linear, or tail) [22]. A single-strain challenge therefore cannot stand in for the product. On RTE meats as a class, a scoping review of 74 studies found HPP in 29 of them [7]. Below 400 MPa, inactivation was typically below 1 log even with long holds [7]. At 500–600 MPa, reported reductions ran from 1 to 7 log CFU/g within about 5 min [7]. Thirty-one of 74 papers omitted standard deviations; 27 used a single strain; meta-analysis was judged impossible [7]. Those rows are contextually comparable as a reported range. They are not interchangeable measurements of one *Listeria* log. Table 1 therefore lists primary challenges as rows, not as a pooled mean.

3.3 Injury that culture methods miss

Apparent inactivation also depends on how survivors are recovered. Selective agar and immediate sampling miss cells that are damaged rather than unable to form colonies later. In fruit purées treated by HPP, inactivation looked 0.9- to 4.5-fold higher in apple than in banana–apple on matched organisms [13]. Immediate reductions on non-selective media in banana–apple were 0.3 log for *Listeria*, 1.4 log for *Salmonella*, and 0.9 log for *E. coli* [13]. Acid exposure and delayed sampling changed those numbers [13]. A paper that plates only on a selective *Listeria* agar the same afternoon as treatment is not measuring the same residual as one that recovers injured cells. Narrative reviews of nonthermal processing collect the same warning: culture reductions can coexist with injured survivors, later outgrowth, and viable but nonculturable cells that still stain as metabolically active [5]. No colony on a selective plate is therefore not automatically death.

Membrane repair can occur on a different clock from colony formation. After 400 MPa, *L. monocytogenes* pore radii estimated from a fluorescence diffusion model shrank from a mean 1.338 nm on day 0 to 0.809 nm on day 4 in the least-damaged cluster [12]. Viable counts had already fallen 7.79 log CFU/ml to below the limit of quantification and stayed there in that experiment [12]. Injury and death can travel together. They should not be collapsed into one log number copied from a graph.

Taken together, Table 1 supports a working rule rather than a pooled D-value. 500–600 MPa for a few minutes is the industrial vegetative window in these rows, and ice, lactate, fat, °Brix,

agar, and strain cocktail are coefficients [2,7,13,18,22]. A 5-log pasteurisation claim cannot be read off that gauge. It has to be compared with a legal or HACCP number.

Table 1. Primary HPP or HPP-hurdle microbial outcomes in the included studies (not pooled).

Paper	Matrix / organism	P/t (and extra hurdle)	Endpoint	Result
Bulut 2021 [18]	<i>E. coli</i> K12 in PBS	400 MPa / 9 min / 4 °C; ± freeze	log CFU·ml ⁻¹	1.83 (4 °C hold) vs 5.63 (-24 °C) vs 6.83 (-80 °C)
Bulut 2021 [18]	<i>E. coli</i> K12 in orange juice	250 MPa / 15 min; ± freeze	log CFU·ml ⁻¹	≤0.42 unfrozen pH 4.5–5.8; 4.15–4.88 when frozen at -80 °C
Serra-Castelló 2021 [22]	<i>L. monocytogenes</i> in cooked ham	HPP 10 min + 0–2.8% lactate	log reduction vs control	0.5–1.46 log less kill at 1.4% lactate
Torrents-Masoliver 2024 [13]	<i>Listeria</i> , <i>Salmonella</i> , <i>E. coli</i> in purées	HPP (see paper)	immediate non-selective log	banana–apple 0.3 / 1.4 / 0.9; apple higher by up to 4.5-fold
Rodríguez 2024 [17]	<i>Cronobacter sakazakii</i> (buffer)	300–400 MPa ± ultrasound	log CFU ml ⁻¹	4.0 log at 300 MPa / 10 min; no colonies at 400 MPa / 3 min or with US+HPP
Nikparvar 2021 [12]	<i>L. monocytogenes</i>	HPP (see paper)	viable count; pore radius	7.79 log to below LOQ; pores shrank over 4 days
González-Cantillo 2026 [23]	lamb burgers	long hold vs two short HHP cycles	mesophiles, <i>E. coli</i>	mesophiles 7 log to 3.7–4.5; <i>E. coli</i> to <1 log
Pimsannil 2025 [24]	precooked crab + chitooligosaccharide	100–300 MPa / 1.5 min	APC, <i>Vibrio</i>	100 MPa ~0.3–0.4 log APC; 300 MPa removed several groups
Guzel 2026 [7]	RTE meats / <i>L. monocytogenes</i>	mostly 300–600 MPa (29 studies)	reported log range	<1 log below 400 MPa; 1–7 log at 500–600 MPa

Primary endpoints in Table 1 are summarised from the notes; recovery methods differ, so rows are not interchangeable [7,13].

4. Pasteurisation performance criteria for milk and juices

That gauge still has to be compared with a legal or HACCP number. Milk is where the law already names the comparison: time, temperature, and a 5 log₁₀ performance criterion.

4.1 HPP milk versus legal heat pasteurisation

Legal milk pasteurisation already names the comparator: time, temperature, and a 5 log₁₀ performance criterion. EFSA modelled industrial 600 MPa for 6 min as the stringent HPP case and still found fewer log₁₀ reductions than legal heat for the pathogens that could be compared [2]. HPP milk was therefore not equivalent to pasteurisation on that metric. Cited *L. monocytogenes* data in milks show the pressure staircase behind the model: 200–250 MPa for 5–15 min often <0.5 log₁₀; 300, 350, and 400 MPa for 5 min about 1, 4.4, and 6.3 log₁₀ [2]. *S. aureus* is repeatedly the more resistant vegetative pathogen under pressure [2,25]. Alkaline phosphatase does not verify HPP the way it verifies heat [2].

The same legal question can be asked with a second endpoint: named pathogens and native bioactive proteins on the same milk. A systematic review of raw bovine milk found 65 papers with pathogen outcomes and 24 with bioactive proteins, but only four papers measured both [25]. Pressure was the strongest statistical contributor to log reductions for *L. monocytogenes*, *E. coli*, and *S. aureus* [25]. For *L. monocytogenes*, treatments above 551 MPa still needed more than about 5 min at 20–30 °C to reach >5 log in the papers that got there [25]. Come-up and come-down alone were not enough. *S. aureus* kills above 5 log at 401–600 MPa often needed long holds. Some of those >5-log *S. aureus* results used 60 °C and are no longer nonthermal [25]. A cycle mild enough to keep immunoglobulin G and lactoferrin is therefore not automatically a 5-log pathogen cycle. The experiments that would show both on the same batch are scarce [25]. A paired high-temperature short-time versus HPP bovine-milk trial that reports both endpoints on the same batches is not among the included studies.

That hole has not stopped narrative dairy reviews from calling 400–600 MPa “sufficient” for microbial control [26] and ultrasound, pulsed electric fields, and supercritical carbon dioxide promising dairy tools [27]. Those sentences compile selected rows. They do not rebut the EFSA model. Table 2 marks the difference.

Primary cream and donkey-milk experiments are consistent with “pressure reduces some flora” rather than with legal equivalence. In dairy cream, 450 MPa for 5 min left total aerobic psychrotrophs similar to raw immediately, with spoilage (>6 log) from day 18 [15]. 600 MPa slowed recovery, but heat still produced a different trajectory [15]. In donkey milk, 200, 400, and 500 MPa cut total aerobic bacteria by 0.85, 1.58, and 4.04 log CFU ml⁻¹, while 75 °C for 2 min cut 1.34 log [9]. Lysozyme loss was much larger after heat in the results section than in the abstract [9]. Read from the results section, 400 MPa is closer to a quality-preserving analogue than 500 MPa is. Neither arm is a named-pathogen performance criterion [9].

4.2 Juice APC is not a named-pathogen 5-log

Juice processors face the same 5-log number, now written for a named pathogen rather than for heat-pasteurised milk. Acid already suppresses many organisms, so a drop in background aerobes is not the same measurement. Whole Concord grape purée showed about 5-log reduction of fresh APC after both HPP and heat [11]. Yeasts and moulds were <1.0 log CFU/g after HPP versus 1.22 after heat, and HPP held those groups down longer in four-month storage [11]. That is a 5-log APC result, not a named *E. coli* O157:H7 result.

Refrigerated stability and enzyme survival can move independently of that APC number. Celery juices after HPP kept total aerobic counts <2 log CFU/ml for 8 weeks, with yeasts and moulds undetectable, while untreated juices spoiled in a week [28]. Semi-industrial red-globe grape juice lost about 3 log aerobic counts after HPP, ohmic heating, or thermal pasteurisation [8]. PPO activity told a different story: about 30% residual after HPP versus >90% inactivation by ohmic heating and by heat [8]. A flow-through dual-frequency sonicator at 50–55 °C reached >3.6, 2.7, and 2.8 log reductions of yeast, mould, and aerobes in apple juice after 450 s efficient time [3]. It still missed a 5-log pathogen pasteurisation criterion, and there was no isothermal heat-only arm [3]. None of the included juice papers reports an inoculated *E. coli* O157:H7 or *Salmonella* 5-log HPP validation.

Table 2 is therefore a scorecard of those benchmarks, not a verdict that “HPP works.”

Table 2. Pasteurisation or 5-log benchmarks versus what included studies actually measured.

Paper	Product	Benchmark discussed	What was measured	Met?
Koutsoumanis 2022 [2]	ruminant milk/colostrum	legal heat pasteurisation / 5 log ₁₀ PC	modelled HPP vs heat log ₁₀	HPP 600 MPa / 6 min < heat for compared pathogens
Sykora 2026 [25]	raw bovine milk	>5 log named pathogens + native proteins	65 microbial + 24 protein papers; 4 overlap	joint optimisation not established
Agriopoulou 2023 [26]	milk products	400–600 MPa “sufficient”	narrative table of cited 5-log rows	cannot replace [2]
Machado 2023 [15]	dairy cream	spoilage delay vs heat	TAP/LAB over 51 days	450 MPa ≈ raw; 600 MPa slower spoilage
Koker 2023 [9]	donkey milk	TABC vs 75 °C; lysozyme/lactoferrin	factorial HHP	400 MPa ~1.6 log TABC; not a pathogen PC
Li 2021 [11]		5-log		

	Concord grape purée		APC and yeasts/ moulds	5-log APC yes; named pathogens not tested
Le 2023 [28]	celery juices	refrigerated stability	TAC, yeasts/ moulds, PPO/POD	TAC held 8 weeks; not a 5-log pathogen test
Eran Nagar 2026 [8]	grape juice	industrial pasteurisation	aerobic, Y&M, PPO	~3 log aerobic; PPO poorly inactivated by HPP
Maghami 2026 [3]	apple juice (sonication)	FDA 5-log pathogen	native yeast/mould/ aerobes	2.7–3.6 log; 5-log not met

5. Spores, heat under pressure, and combined hurdles

Those juice and milk rows are still vegetative stories. Industrial HPP at ambient or chill temperature is not a botulinum cook.

5.1 Spores, adiabatic heat, and refrigeration

Pressures below 600 MPa at room temperature struggle with bacterial spores because the dormant core is dehydrated and baroprotective [1]. Some *Bacillus subtilis* spores have been reported to survive 1200 MPa at ambient temperature in laboratory equipment [1]. HPTP uses the adiabatic temperature rise to add heat without a long retort. That rise is about 3 °C per 100 MPa in water-rich foods, about 4.5–6.3 °C per 100 MPa in chicken and beef fat, and still higher in oils [16]. Hold and decompression cooling then follow. Time spent above 100 °C can therefore be much shorter than a retort, which is why a baby-food purée can be discussed as an F0 equivalent of minutes rather than a >60 min cook [16].

Commercial HPP machines usually do not supply that heat. Dedicated industrial HPTP vessels were still not a retail line in the included engineering review [16]. Packaging must survive both 600 MPa and about 120 °C [4,16]. Refrigerated distribution below 7 °C is therefore the practical control on spore outgrowth after a cold 600 MPa cycle [4,16]. Processors sell HPP as a vegetative refrigerated hurdle rather than as a botulinum cook. The physics of HPTP is not unknown. The equipment and packaging have not made that physics cheap.

Those industrial limits come with kinetic parameters, not a new challenge trial [4]. Vegetative inactivation in juices is often first-order, with D-values at 350 MPa on the order of 0.6–2 min for some lactic acid bacteria and yeasts, because those cells are already pressure-sensitive in acid, high-aw matrices [4]. Spore curves in fruit products are often concave-up (Weibull or biphasic) with tails. As time increases, remaining spores look harder. A single ambient-600 MPa log therefore cannot be extrapolated to a 5–6 log pasteurisation target [1,4]. For

Alicyclobacillus acidoterrestris spores, a cited D-value at 600 MPa and 45 °C was 8.6 min in apple juice, 20 min in 20 °Brix lime concentrate, and 46 min in 30 °Brix blackcurrant concentrate [4]. Soluble solids protect spores the way fat and lactate protect vegetative cells. *Neosartorya fischeri* ascospores in apple juice illustrate the same point with heat under pressure: 600 MPa at 75 °C for 10 min gave 3.3 log cfu/ml versus no reduction at 75 °C without pressure, in work cited by the fruit HPTP review [1]. That is synergy. It is still not a 6-log pasteurisation of the most resistant mould spore in every concentrate.

Water activity can erase even that synergy, because pressure needs a compressible aqueous phase in which to act. In undiluted honey (aw about 0.55–0.57), pressure-assisted thermal processing at 600 MPa / 15 min / 85 °C and heat alone at 85 °C / 15 min both failed to reduce inoculated *B. subtilis* TEL spores, including after 24 h [29]. Diluting to aw 0.85 or 0.90 restored PATP lethality at 600 MPa / 15 min / 75 °C [29]. Room-temperature 600 MPa still did not cut endospore loads [29]. Low-aw foods can therefore make “HPP plus heat” look like heat, which already failed. PATP of honey is discussed for diluted pharmaceutical preparations rather than as a way to pasteurise table honey without changing the product [29].

5.2 Vegetative hurdles are not a spore cook

If even heat under pressure can fail when water activity is low, milder nonthermal partners cannot be sold as a spore cook. They are used to finish vegetative kill at milder set-points. Plant-food combination work starts from pressure-resistant strains and from storage regrowth after HHP alone, and treats nonthermal partners as process-design problems rather than as seasoning [14]. Blueberry juice thermosonication plus 400 or 600 MPa / 5 min is recited among combinations that let each factor run milder [21]. Ultrasound plus HPP eliminated detectable *C. sakazakii* in buffer at conditions where 300 MPa / 10 min still left about 2 log [17]. Freeze-then-pressure did the same job for *E. coli* K12 [18]. None of those combinations is a spore sterility proof.

Table 3 records spore and hurdle rows that can be cited without pretending they share a D-value.

Table 3. Spore and hurdle outcomes (reviews cited as such).

Paper	System	Process	Result
Zhang 2022 [1]	fruit/vegetable spores (review)	HPTP vs HPP vs heat	HPP ambient insufficient for many spores; example 3.3 log <i>N. fischeri</i> at 600 MPa / 75 °C / 10 min (cited)
Houska 2022 [4]	juices (review)	600 MPa ± heat	<i>A. acidoterrestris</i> D_45°C,600 MPa 8.6 min apple vs 46 min 30 °Brix concentrate (cited)

Scepankova 2022 [29]	<i>B. subtilis</i> in honey	600 MPa / 85 °C / 15 min	no reduction in undiluted honey; kill after dilution to aw 0.85–0.90
Knoerzer 2025 [16]	low-acid foods (review)	HPTP vs retort	adiabatic heating shortens time above 100 °C vs >60 min retort
Xia 2022 [14]	plant foods (mini-review)	HHP + nonthermal hurdles	HHP alone not complete; injured-cell regrowth
Masilamani 2025 [21]	mixed foods (review)	nonthermal hurdles	combinations allow milder intensities (recited examples)
Rodríguez 2024 [17]	<i>C. sakazakii</i> buffer	US + HPP	no detectable CFU with combination
Bulut 2021 [18]	<i>E. coli</i> K12	freeze + HHP	freeze converts <2 log to 5–7 log in PBS

6. Quality endpoints that do not move with plate counts

A clean plate still does not make the food “fresh-like.” That claim has to name an enzyme, a protein, or a texture.

6.1 Enzymes and bioactive proteins

Enzyme activity and plate count are different targets on the same cycle. PPO and peroxidase in celery juice fell after HPP and stayed low while microbial counts held for 8 weeks [28]. In grape juice, the same class of enzyme barely moved under HPP, while heat and ohmic heating removed more than 90% of PPO [8]. A cycle that already looks adequate on aerobic plates can therefore leave the browning catalyst in place. Cloud-stable orange juice historically uses 90–99 °C for 15–30 s to hit thermostable pectin methylesterase, because that fraction is the cloud-collapse enzyme [4]. Some pressure studies inactivate the heat-labile PME fraction at ≥500 MPa, with large residual variation by cultivar [4]. Enzyme and microbe remain different specifications.

Native milk proteins and plant allergens are another protein class. They unfold under pressure because non-covalent bonds yield and hydrophobic interiors become exposed [6]. A protein-source review reports lower allergenicity and higher digestibility or bioactivity for cereal, legume, and nut proteins after HHP, higher surface hydrophobicity for some seed and vegetable proteins, and mixed foaming, gelling, and antioxidant shifts in dairy [6]. That is a functional-food story, not a *Listeria* story, and it is qualitative across methods. Holding IgG and lactoferrin is why some human-milk and bovine-milk arguments prefer HPP over Holder pasteurisation or high-temperature short-time: heat denatures those proteins more readily than a moderate pressure cycle does. The four overlap papers that measured both pathogens and proteins are not enough to set a dual specification with the pathogen logs already shown for

milk [25]. The cycle that keeps the protein is not automatically the cycle that delivers 5 log of *S. aureus*. Donkey milk shows the trade in one factorial: 500 MPa maximised aerobic kill and maximised lactoferrin activity loss [9]. Choosing the microbial maximum spent the quality claim that had justified avoiding 75 °C.

6.2 Texture when 600 MPa is not gentle

Texture is the next named endpoint, and it can fail while the plate looks clean. Unwashed *Saccharina latissima* hardness fell from 312 g untreated to 38 g at 600 MPa (about 88% drop), with compression distance halved [10]. That kelp is not a ready-to-eat meat. It is a reminder that 600 MPa is not gentle for every gel. Macroalgae-fortified sausages gained minerals and, immediately after HPP, total aerobic mesophiles <1.0 log CFU/g in the fortified HPP batch [30]. Pumpkin pulp with a natural preservative plus HPP showed no microbial growth for 45 days, with better colour and firmness than a thermal comparator in the NBP formulations [31]. Minimally processed potatoes lost lightness and gained ΔE after HPP or HPP/ultrasound [32]. Enterobacteriaceae became undetectable after HPP, while combined ultrasound did not stay different from untreated APC by day 14 [32]. Quality and kill can trade in either direction.

6.3 Chilled seafood and spoilage flora

Chilled seafood is where that trade is sold: histamine, *Vibrio*, and *Pseudomonas*, not a 5-log *Salmonella* graph. In a dual-species fillet trial, treated salmon and plaice stayed acceptable to day 12 while untreated flora ran ahead, and histamine stayed low [33]. Mackerel fish balls at 250–350 MPa started <1 log CFU/g versus about 2.2 in controls and stayed down for 7 days at 4 °C [34]. Counts can look similar to a 121 °C retort at day 7, while the thermal product is a different food. Mint extract plus 400 MPa on squid delayed total viable count relative to pressure or extract alone [35]. Several treatments still crossed a TVC limit by day 6 without the mint. The extra kill is modest, so the benefit is days of delay rather than a spore cook. Precooked crab needed 300 MPa more than 100 MPa to move *Vibrio* and hydrogen-sulphide producers [24].

Surface geometry and formulation can cap even those storage gains. Low-temperature high-pressure carbon dioxide at 6.5 MPa / 10 °C / 15 min inactivated 99.45% of *E. coli* in suspension but only 93% on prawn surfaces (67% for *S. aureus* on prawn) [36]. Topography limited a process that looked excellent in buffer. Lamb burgers with lupin flour dropped coliforms and *E. coli* below 1 log whether a long hold or two short HHP cycles were used [23]. Sous-vide products after high hydrostatic pressure stayed below 1.30 log CFU/g on TSA through day 45 in one trial, while controls exceeded 5 log [37]. An aquatic-food UHP review collects these patterns without a pooled shelf-life [38]. Ultrasound-assisted ultra-high pressure

in a lactic fermentation changed antioxidant readings and sugar consumption at 24–40 h [39]. That is process intensification of a ferment, not pasteurisation of a pathogen. Mineral-fortified sausages and pumpkin pulp with a natural preservative show HPP as a shelf-life tool in formulated products where “fresh-like” is already compromised by other ingredients [30,31].

Taken together, HPP quality is sometimes better than heat (grape APC storage [11], donkey-milk enzymes at 400 MPa [9], avocado colour in commercial examples reviewed for plant foods [4]) and sometimes worse or simply different (kelp hardness [10], grape PPO [8], cream at 450 MPa [15]). Plate counts do not arbitrate those sentences, and they also do not license treating ultrasound, plasma, or irradiation as interchangeable with pressure.

7. Other nonthermal fields in the same literature

Those methods appear here as supporting technologies, not as a balanced map of the nonthermal field.

7.1 Ultrasound, plasma, and irradiation as different unit operations

Among those supporting technologies, the sonicator in apple juice is the clearest included example: dual-frequency excitation and a Venturi at reduced temperature reached a few log of spoilage flora and improved visual sedimentation, with a specific energy of 0.044 kWh/L at the best condition, and did not replace a 5-log pathogen pasteuriser [3]. Ultrasound reviews of mycotoxin and fungal decontamination quote high percentage degradations of aflatoxin B1 under long, high-intensity treatments that are not juice HACCP validations [40]. Sugarcane-juice comparisons of ultrasound, pulsed electric fields, HPP, and microfluidization versus heat treat HPP as the most used nonthermal option in that matrix [41].

Plasma is a surface technology, so a log in buffer or on a fillet does not transfer to a wash line. Gliding-arc plasma for 5 min plus C-phycocyanin delayed rainbow trout total viable count crossing 6 log CFU/g from day 3 (control) to day 12, and kept total volatile nitrogen below 30 mg N/100 g through day 18, while untreated fillets did not [19]. Plasma-functionalised water on fresh-cut lettuce reached 4–5 log₁₀ laboratory reductions in multi-step washing and about 2 log₁₀ in a pilot scenario that resembled a factory wash [20]. Scale, not species, is the disagreement.

Ionising irradiation of fruit and vegetables is summarised with the FAO/IAEA/WHO statement of no toxicological problem up to 10 kGy and an FDA refrigerated-meat limit of 4.5 kGy, plus typical lethal-dose tables [42]. That is a different legal history from HPP and should not be blended into a single “nonthermal efficacy.”

Broad matrix reviews rank HPP among options with typical vegetative reductions of 3–6 log₁₀ and a high technology-readiness level, while insisting that no method is universal [43]. Beverage reviews argue that nonthermal methods can match thermal preservation with better nutrient retention, reciting a soy smoothie at 650 MPa / 20 °C / 3 min among examples [44]. Those sentences are maps. They do not override Tables 1–3. Ranking fields by typical log₁₀ and readiness is useful for an engineer choosing a vessel [43]. It is not a substitute for the milk opinion or the RTE-meat scoping range [2,7]. The smoothie example sits at 650 MPa, which is already above many commercial 600 MPa ceilings discussed for plant foods [4,44]. Mycotoxin percentages come from long ultrasonic treatments that would not be confused with a 3-minute HPP juice cycle if the methods were read [3,40]. Ranking of HHP versus ohmic heating in sugarcane juice is matrix-bound [41]. Ultrasound-assisted ultra-high pressure fermentation is a different unit operation: antioxidants and sugar use in a lactic culture, not a pathogen specification [39]. Folding those rows into Tables 1–2 would mix unit operations that share a label more than a mechanism.

The physical picture is still pressure for HPP, cavitation for the sonicator, reactive species for plasma, and photons or electrons for irradiation. A combined factory line can use more than one, but a review that averages their log reductions would hide the scale-up failures already visible for plasma water and the 5-log miss for reduced-temperature sonication [3,20].

Discussion

The included literature supports two conclusions that look opposed only if they are forced into one sentence. At 500–600 MPa for a few minutes, HPP repeatedly reduces vegetative flora in refrigerated juices, guacamole-style herb pastes, and post-lethality RTE meats [4,7]. Those products can avoid a second cook while a cold chain remains part of the labelled life. On milk, the same window is not equivalent to legal pasteurisation on EFSA's comparison. Indicator enzymes do not transfer. *S. aureus* is repeatedly the more resistant vegetative pathogen. Bioactive-protein preservation has not been co-validated with 5-log pathogen kill on enough shared batches [2,25]. Those are different questions: a commercial vegetative hurdle versus legal heat equivalence. Collapsing them into “HPP is safe and mild” is not licensed by the measurements.

The spread of reported logs is information about coefficients, not a reason to average them. A scoping review of RTE meats refused meta-analysis because variance, D-value-only papers, and single-strain challenges broke the table [7]. Fruit composition and agar changed apparent kill [13]. Lactate, a normal ham ingredient, subtracted kill [22]. Injury and viable-but-nonculturable states can coexist with culture success [5]. Averaging 1 log and 7 log at “600 MPa” is averaging different foods, strains, and methods.

Quality claims need the same refusal. PPO, lysozyme, kelp hardness, and APC do not move together. Grape juice showed similar microbial drops and unlike enzyme drops across HPP, ohmic, and thermal treatments [8]. Allergenicity trends after HPP are a further endpoint, still largely unconnected to the *Listeria* rows [6]. A cycle can therefore be adequate on one specification and inadequate on another. The specification has to be named.

Taken together, lethality, legal status, and quality are not properties of a megapascal value. They emerge from the interaction of pressure with ice, solute, pH, water activity, strain, recovery medium, and the endpoint that was actually assayed. That is why an RTE-meat success, an EFSA milk non-equivalence, and a grape-juice APC result can all be true without licensing a single industrial claim.

Ultrasound, plasma, and irradiation remain too thinly evidenced here to rank against HPP as substitutes. The included sonicator missed a 5-log pathogen criterion in apple juice [3]. Plasma-functionalised water lost about 2–3 log moving from bench lettuce to a pilot wash [20]. Irradiation evidence here is a mini-review with a different legal history [42]. High-pressure carbon dioxide looked strong in suspension and weaker on prawn [36]. Substituting “nonthermal” for “HPP” in a conclusion would erase those differences of mechanism, scale, and regulation.

Even inside HPP, pH still splits the safety class. Low-acid vegetable juices (pH above 4.6) remain a botulinum problem if the cold chain fails [4]. Processors still have to show that the product cannot support growth and toxin within its labelled life. Refrigeration after HPP is therefore not only a quality choice. It is the control that makes commercial 600 MPa cycles legal for soups and some vegetable juices that would otherwise need a full sterilisation mentality [4,16]. Fruit juices below pH 4.6 are kinder to vegetative control, but acid-tolerant *E. coli* O157:H7 and *Salmonella* can survive for weeks at low pH with a low infectious dose [4]. A named-pathogen 5-log test is not optional on that reading. The included grape and celery papers did not run that test [11,28].

What cannot currently be concluded follows from those holes rather than from a generic call for more work. Named-pathogen juice validations (*E. coli* O157:H7 or *Salmonella* under HPP) and a paired bovine-milk high-temperature short-time versus HPP trial with both pathogen and protein endpoints are not among the included full texts. Abstracts can disagree with results, as in donkey milk [9]. Several reviews recycle the same spore examples. Honey, kelp, crab, and cream are not one population. None of the included primary papers is a randomised trial of human foodborne disease. The plant-food kinetics review adds D-values and equipment limits; it does not add a new inoculation [4]. Those limits do not weaken the vegetative-hurdle claim. They bound it.

What should be measured next follows from the same bounds. A bovine-milk HPP versus high-temperature short-time experiment that reports the same pathogen cocktail (including *S. aureus*) and the same bioactive proteins on each batch, with recovery media stated, would close the dual-specification gap. Juice HPP validations still need inoculated *E. coli* O157:H7 or *Salmonella* against the FDA 5-log criterion rather than APC. Spore work needs measured temperature under pressure, °Brix, and a Weibull or biphasic fit instead of a single ambient-600 MPa log. RTE meat challenges that use strain cocktails, report standard deviations, and treat lactate or fat as designed factors would make the 1–7 log window interpretable [7,22]. Plasma and ultrasound still need paired bench-versus-plant measurements, given the lettuce pilot drop [20] and the sonicator's missing heat control [3].

Conclusions

HPP reduces many vegetative bacteria in high-water-activity foods at 500–600 MPa, and that reduction is large enough, in the right matrix, to support refrigerated juices and post-lethality meats. The reduction is not a constant, and it is not a spore kill. A cycle that looks adequate on non-selective agar the same day may look weaker after acid recovery, in a high-lactate ham, or after a week of chilled storage [13,15,22].

Legal milk pasteurisation is not interchangeable with 400–600 MPa on the included opinion: the industrial cycle underperforms heat on comparable log₁₀ reductions [2]. A 5-log APC in grape purée is likewise not a juice pathogen validation [11]. Spore tails remain, so either HPTP or a cold chain has to be planned, with water activity as a designed factor [1,4,29]. Enzymes, bioactive proteins, and texture have to be measured on the same samples as microbes, because they do not covary [8,9,10]. Recovery methods and strain sets can move a log reduction as much as 100 MPa can, and they should be reported with the set-point [7,13]. Ultrasound, plasma, and irradiation stay in separate sentences until their included primary data are as dense as the HPP rows in Tables 1–3.

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